

6.5.11.4. Subject Key Identifier Extension

The subject key identifier extension provides a means of identifying Matter certificates that contain a particular public key.

When present, the subject key identifier extension SHALL be treated as **non-critical** and it SHALL be marked as **non-critical** in the corresponding X.509 certificate. The **critical** field SHALL NOT be encoded in the Matter certificate structure.

The context-specific tag number **subject-key-id** [4] SHALL be used to identify a subject key identifier extension entry in the Matter certificate **extensions** list.

The Subject Key Identifier field SHALL be derived from the public key using method (1) described in section 4.2.1.2 of [RFC 5280]. Thus, the **subject-key-id** SHALL be composed of the 160-bit SHA-1 hash of the certificate's subject public key value.

See [Section 6.1.2, “Key Identifier Extension Constraints”](#) for additional constraints.

6.5.11.5. Authority Key Identifier Extension

The authority key identifier extension provides a means of identifying the public key corresponding to the private key used to sign a Matter certificate.

When present, the authority key identifier extension SHALL be treated as **non-critical** and it SHALL be marked as **non-critical** in the corresponding X.509 certificate. The **critical** field SHALL NOT be encoded in the Matter certificate structure.

The context-specific tag number **authority-key-id** [5] SHALL be used to identify an authority key identifier extension entry in the Matter certificate **extensions** list.

Note that the authority key identifier extension field in an X.509 certificate may optionally include issuer and serial number fields, which are not supported by Matter certificates.

The Authority Key Identifier field SHALL be derived from the public key using method (1) described in section 4.2.1.2 of [RFC 5280]. Thus, the **authority-key-id** SHALL be composed of the 160-bit SHA-1 hash of the public key used to verify the certificate's signature.

See [Section 6.1.2, “Key Identifier Extension Constraints”](#) for additional constraints.

6.5.11.6. Future Extension

The Matter certificate is designed with extensibility in mind and this field is added to support arbitrary certificate extension in the future.

Note that implementations that do not support specific future extension will ignore it but will be able to use it for the Matter certificate signature validation. If ignored extension is marked as critical then validation of the corresponding Matter certificate SHALL fail.

The context-specific tag number **future-extension** [6] SHALL be used to identify all future extension entries in the Matter certificate **extensions** list. There MAY be more than one future extension field.